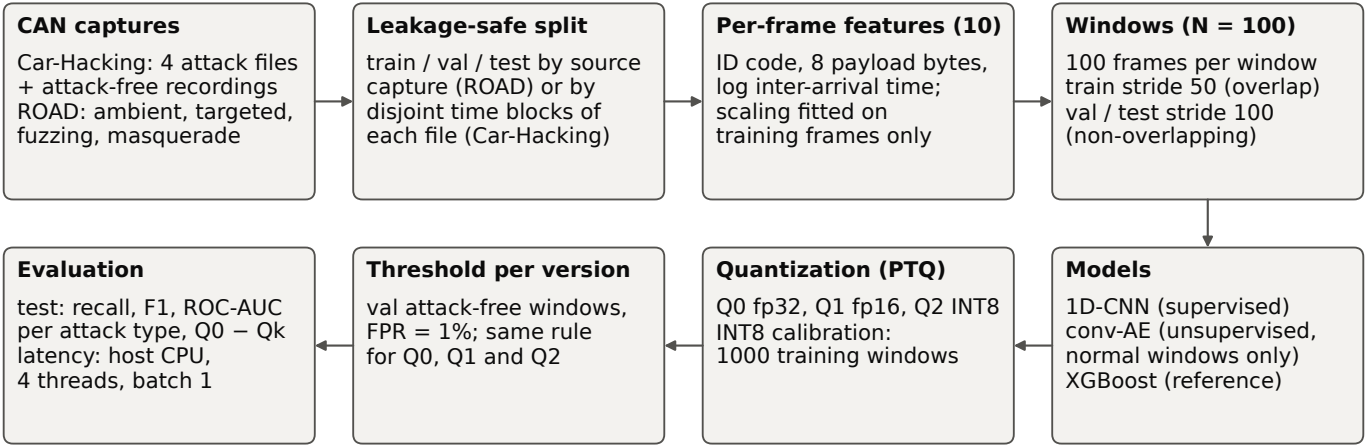
CAN captures
Car-Hacking: 4 attack files + attack-free recordings
ROAD: ambient, targeted, fuzzing, masquerade
Leakage-safe split
train / val / test by source capture (ROAD) or by disjoint time blocks of each file (Car-Hacking)
Per-frame features (10)
ID code, 8 payload bytes, log inter-arrival time; scaling fitted on training frames only
Windows (N = 100)
100 frames per window
train stride 50 (overlap)
val / test stride 100 (non-overlapping)
Models
1D-CNN (supervised)
conv-AE (unsupervised, normal windows only)
XGBoost (reference)
Quantization (PTQ)
Q0 fp32, Q1 fp16, Q2 INT8
INT8 calibration: 1000 training windows
Threshold per version
val attack-free windows, FPR = 1%; same rule for Q0, Q1 and Q2
Evaluation
test: recall, F1, ROC-AUC per attack type, Q0 − Qk
latency: host CPU, 4 threads, batch 1